

INTRODUCTION TO FEW-SHOT ANOMALY DETECTION

- Few-shot anomaly detection in time-series data focuses on identifying unusual patterns in sequential data such as stock prices, sensor readings, or website traffic. Instead of relying on large amounts of labeled data, few-shot learning enables the system to learn normal behavior from a very small number of examples. This makes it especially useful in real-world situations where anomaly data is rare or difficult to collect. By detecting deviations from normal patterns without requiring extensive labeled datasets, the system can effectively identify potential issues at an early stage. As a result, it plays a crucial role in the early detection of equipment failures, financial fraud, security breaches, and other system faults, helping organizations respond quickly and reduce potential risks.

Challenges in Time-Series Anomaly Detection

- Anomalies are rare and hard to identify.
- Lack of labeled data and high cost of manual labeling.
- Time-series complexity (trend, seasonality, noise).
- Traditional methods require large datasets.
- Need for adaptable and data-efficient models.



FewSOME & Siamese Network Approach

- **FewSOME: One-class few-shot anomaly detection method.**
- **Uses Siamese Networks with shared weights.**
- **Learns feature embeddings of normal data.**
- **Introduces Stop Loss to prevent representational collapse.**
- **Outperforms traditional methods even with very few samples.**



Time-Series & Industrial Applications

- Domain adaptation with LSTM autoencoders for HVAC systems (MZVAV-2 dataset).
- Achieved strong performance (AUC = 0.873).
- TAMA: Uses large multimodal models (LMMs), converts time-series to images.
- FastRecon: Feature reconstruction-based industrial anomaly detection.
- MOSAD: Open-set anomaly detection detecting both seen and unseen anomalies.



Conclusion & Future Directions

- Few-shot anomaly detection is effective when data is scarce.
- Methods like FewSOME, FastRecon, and MOSAD show strong results.
- Emerging trends include multimodal models and domain adaptation.
- Future work: Improve scalability and interpretability.
- Highly valuable for real-world industrial and monitoring systems.

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Advanced Few-Shot Anomaly Detection in Time-Series

- Combine Self-Supervised Learning + Transformer + Siamese Network
- Detect anomalies with very limited labeled data
- Improve generalization for unseen anomalies
- Add explainability for real-world trust

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Proposed Hybrid Architecture

Model Components:

- ◆ LSTM → Capture temporal patterns
- ◆ Transformer → Handle long-range dependencies
- ◆ Siamese Network → Few-shot similarity learning
- ◆ Distance-based anomaly scoring

Expected Advantage:

- Better feature extraction
- Robust performance with small datasets
- Improved anomaly discrimination

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Self-Supervised Pretraining Strategy

Step 1: Pretraining

- Masked Time-Series Modeling
- Contrastive Learning approach

Step 2: Few-Shot Fine-Tuning

- Use small normal dataset
- Adapt to target domain

Benefit:

- Reduces dependency on labeled data
- Learns strong representations

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Explainability & Open-Set Adaptation

Explainability Module:

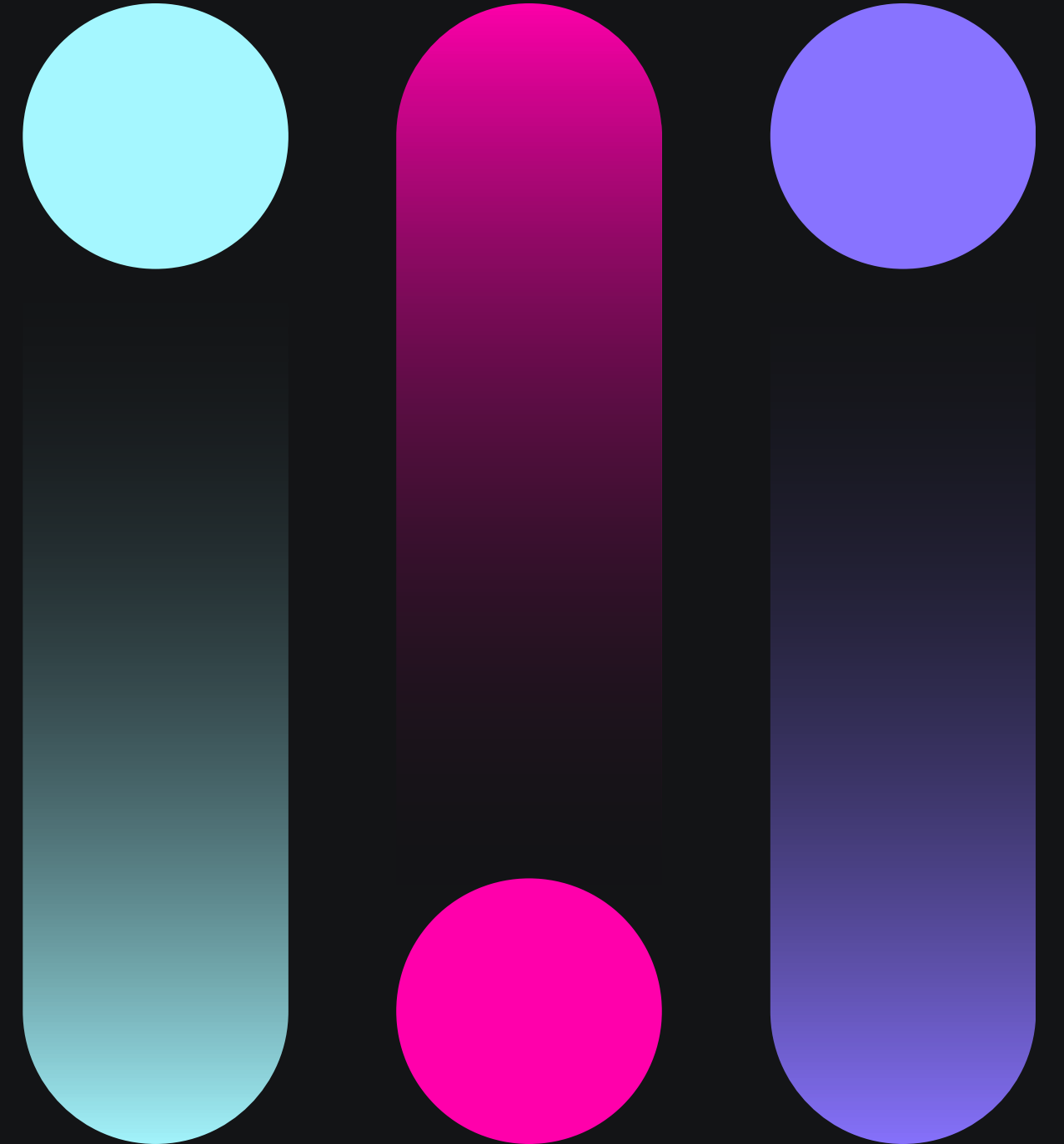
- Attention visualization
- Important time-step highlighting
- Anomaly contribution scoring

Open-Set Detection:

- Adaptive threshold
- Memory bank for unseen anomaly patterns

Outcome:

- Detect both seen & unseen anomalies
- Increase industry trust



Expected Impact & Future Scope

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Expected Results:

- Higher AUC & F1 Score
- Better performance in small-data scenarios
- Real-time anomaly monitoring capability

Future Scope:

- Edge device deployment
- Industrial IoT integration
- Healthcare monitoring systems



Thank You

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